Timothy Jin

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EDUCATION

The University of British Columbia | *B.Sc. Computer Science — 3.9 / 4.0 GPA* Sep 2021 – May 2026

Data Structures & Algorithms · Machine Learning · Networking & Security · Databases · Distributed Systems

EXPERIENCE

Amazon | *Software Development Engineer Intern* May 2025 – Aug 2025

- **Eliminated engineer intervention for 80% of tickets** submitted by internal teams requesting help with the Prime Subscriptions API — built an **agentic assistant** in Java, Python, and TypeScript powered by **AWS and Claude LLMs** to automatically analyze user issues, surface insights, and run API tests.
- **Reduced debugging time by 80%** by migrating the **Prime Subscriptions API** from compressed file-based logs (requiring manual grepping) to centralized, queryable **CloudWatch logging** across **9 global deployments**.
- **Reduced manual scaling work by 95%** for high-traffic shopping events (e.g. Prime Day) by integrating the Prime API into an automated capacity planning interface, replacing a process that previously required weeks of manual effort.

Stanford University - Dept. of Emergency Medicine | *Machine Learning Researcher* May 2024 – Oct 2025

- Improved ED patient outcome prediction by **0.12 AUROC** by tuning and deploying **LightGBM** models trained on historical patient data to a hospital dashboard, enabling better resource allocation for **100+ patients daily**.
- Identified the optimal model architecture—**improving accuracy by 15%**—by benchmarking PyTorch transformer and LLM-assisted LightGBM models against baselines across **millions of patient records**.
- Improved ED physician confidence in model decisions by building a **React + AWS dashboard** with **SHAP-based explainability**, making predictions transparent and actionable for physicians.

Rivian | *Software Engineer Intern* May 2024 – Dec 2024

- Built foundational tooling to validate Rivian's transition from **REST/GraphQL**-based communication to **Protocol Buffers**, maintaining **100% feature test coverage** by developing Rivian's first **Python Protocol Buffer pub/sub framework** and NATS client tool for critical simulated cloud-based testing, reducing dependence on limited prototype hardware.
- Proposed and drove development of an **agentic** full-stack vehicle simulator built with **React and AWS** that let **100+ engineers and designers** test mobile and cloud features, eliminating hours of authentication and CLI setup.
- Led mobile integration of the 2024 Halloween feature and multiple other features, shipping to **100,000+ customers** across **7 major app releases** by designing test strategy and driving cross-team alignment.

AWS Cloud Innovation Centre (UBC) | *Software Developer Co-op* Jan 2023 – Aug 2023

- Built a **Flutter/Dart mobile app** with a TypeScript-based **AWS** serverless backend to collect Parkinson's disease symptom data from **thousands of patients annually** with minimal discomfort, supporting clinical research teams.
- Improved balance assessment for rehabilitation patients using **transformer-based ML** on IMU sensor data, delivered via a **React and Flutter UI** with a TypeScript and Python **AWS** backend.
- Presented at the **2024 Interdisciplinary Conference in Rehabilitation Science BC** (INCREASE BC).

PROJECTS

FinHog · *Agentic AI, TypeScript, Python, AWS, React* Sep 2025

- Won **1st place at HackMIT** (1,500 participants) for Anthropic's Best Use of Claude, and **2nd place** for Warp's (YC W23) Best Financial Visualization Agent.
- Built an **autonomous financial analytics agent** that analyzes financial data in real time, automatically generating and adapting visualizations to surface the most meaningful insights — delivered in under 24 hours.

Vancouver Transit App · *Dart, Flutter, iOS, Android* Nov 2019 – Jan 2022

- Built a **Flutter**-based iOS and Android app providing real-time transit data, reaching **1,000+ active users** and **10,000+ installs** across the Apple App Store and Google Play Store.

TECHNICAL SKILLS

Languages	TypeScript, Python, C++, Java, SQL, JavaScript, Dart, GraphQL, HTML/CSS
Frameworks & Libraries	Protocol Buffers, AWS CDK, Flutter, React, Node.js, Pandas, NumPy, JUnit
Tools & Platforms	AWS, PostgreSQL, MySQL, Docker, Git, CI/CD, NoSQL, Xcode, Android Studio, VS Code